

Generative AI Metrics & Insights

1. Introduction

Generative AI metrics evaluate the quality, performance, safety, and business impact of AI-generated outputs.

2. Key Metrics

- Accuracy, Relevance, Coherence, Fluency
- BLEU, ROUGE, METEOR (for NLP)
- Latency, Throughput (Performance)
- Hallucination Rate, Bias Detection (Safety)
- ROI, User Engagement (Business)

3. Advanced Metrics

Perplexity: Measures how well a model predicts text. Lower is better.

Hallucination Rate: Measures incorrect or fabricated outputs.

Groundedness: Measures factual correctness.

Diversity: Measures variation in outputs.

FID (Fréchet Inception Distance): Used to evaluate image generation quality.

$$\text{FID} = \|\mu_r - \mu_g\|^2 + \text{Tr}(\Sigma_r + \Sigma_g - 2(\Sigma_r \Sigma_g)^{1/2})$$

Lower FID indicates better quality.

4. GAN Full Form

GAN = Generative Adversarial Network

Includes Generator and Discriminator models.

5. Real-Time Use Case: AI-Powered Incident Management

AI integrates with monitoring tools to analyze logs, generate incident summaries, and suggest fixes.

Metrics used:

- Accuracy & Relevance – Correct root cause detection
- Hallucination Rate – Avoid false insights
- Latency – Fast response during incidents
- MTTR Reduction – Business impact
- Human Feedback – Engineer validation

Additional Use Case: Healthcare Imaging

AI generates medical images. FID is used to compare real vs generated images.

Lower FID ensures high-quality, reliable medical images.

6. Summary

Metrics ensure quality, performance, safety, and business value in real-world AI systems.